

Total No. of Questions: 10]

SEAT No. :

P2970

[6004]-953

[Total No. of Pages : 2

B.E. (Mechanical Engineering)
ADVANCED MANUFACTURING PROCESSES
(2015 Pattern) (Semester-II) (Elective-IV) (402050A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, & Q.3 or Q.4, & Q.5 or Q.6, & Q.7 or Q.8, & Q.9 or Q.10.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.

Q1) a) Examine with neat sketch, stretch forming and list their applications. **[6]**

b) Explain the construction and working of Ultrasonic welding. **[4]**

OR

Q2) a) Explain with neat sketch, Magnetic pulse forming and list their applications. **[6]**

b) List applications of adhesive bonding. **[4]**

Q3) a) Explain with sketch working principle of Abrasive Water Jet machining with the process parameters. **[6]**

b) Examine the process of underwater welding. **[4]**

OR

Q4) a) Examine working principle of Wire-EDM with the process parameter. **[6]**

b) Write short note on welding of plastics and composites. **[4]**

Q5) a) Examine the process of Focused Ion Beam Machining. **[6]**

b) Explain the challenges in micro and nano fabrication process. **[6]**

c) Write short note on Diamond micro machining. **[4]**

OR

P.T.O.

- Q6)** a) Write in detail the need of micro machining. [6]
b) Examine the process of photochemical machining. [6]
c) Write short note on Lithography. [4]

- Q7)** a) Give classification of additive manufacturing processes. [6]
b) Write note on post processing of parts manufactured by additive manufacturing processes. [6]
c) Write applications of additive manufacturing processes in medical technology. [4]

OR

- Q8)** a) What are factors which play important role while designing the object which is manufactured by additive manufacturing? [6]
b) Explain any one Additive Manufacturing process with its details. [6]
c) Write application of additive manufacturing processes in aerospace industry. [4]

- Q9)** a) Examine operating principal of Scanning Electron Microscopes with neat sketch. [6]
b) Explain in details, the importance of material characterization. [6]
c) Describe in details. the applications of microscopes. [6]

OR

- Q10)** a) Examine operating principal of Energy-dispersive X-ray spectroscopy. [6]
b) Describe in details, the applications of spectroscope. [6]
c) Write short note on material characterization. [6]

